

SUMMARY TABLE: GENERAL SCIENCE GRADE 10

Sub-Strand 2.3: Chemical Bonding — Teacher Reference (Pre-filled)

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Sub-Strand	Sub-Strand 2.3: Chemical Bonding
Driving Question	DRIVING QUESTION: Why do the aluminium sufuria, the graphite pencil and the table salt on our kitchen table look similar as solids but behave so differently — and what is happening between the atoms inside each one that controls every property we can measure?

INSTRUCTIONS

FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 The Three Kitchen Mysteries: Anchoring the Phenomenon & Eliciting Prior Knowledge	Students handled and examined three household objects — an aluminium sufuria (or sufuria lid), a graphite pencil, and a pinch of table salt (NaCl) — and recorded observations about their appearance, texture, hardness, and behaviour when tested for electrical conductivity with a simple circuit board. All three appeared as grey/white solids, yet they responded differently: the sufuria conducted electricity readily; the graphite pencil lead conducted weakly; the salt conducted only when dissolved in water. Teachers should note that at this stage students have NOT yet received instruction — the activity is purely phenomenological. Expect surprise and productive confusion, especially about graphite conducting electricity despite being a non-metal.	Students surfaced and documented prior knowledge and misconceptions about electrical conductivity, melting points, and mechanical properties of common solids. Key baseline ideas that typically emerge include: 'metals conduct, non-metals don't' (a useful but incomplete rule students will refine), 'hard things have high melting points', and 'salt dissolves so it must be soft'. These ideas should be recorded verbatim on the Driving Question Board (DQB) and revisited in every subsequent lesson. Pedagogical note: resist the urge to correct misconceptions now — eliciting them authentically is the lesson's core purpose.	No formal explanation is constructed in Lesson 1. Instead, the teacher anchors the entire unit by framing the three objects as an unresolved mystery: 'We will spend the next eight lessons building the explanation from the inside out — starting with the atoms themselves.' The teacher introduces the DQB and records the class's best current explanations alongside their questions. Students write an individual 'prediction letter' describing what they think is happening inside each solid; these are sealed and returned in Lesson 8 for self-assessment. Pedagogical note: the prediction letter is a metacognitive anchor — it makes prior knowledge visible and creates a personal stake in the unit's resolution.
Lesson 2 Predicting the Invisible: What Are Our Three Kitchen Items Made Of?	Students retrieved and reviewed the periodic table, focusing on Group 1, Group 6/7, and Group 13 elements relevant to the three kitchen items (Na, Cl, Al, C). Using electron configuration cards (or a projected periodic table), they identified the number of	Learners activated prior knowledge of valence electrons and electron arrangement from Sub-Strand 2.1. They practised writing structured scientific predictions using the 'I predict... because... and I expect to see evidence of this when...' framework — an	Students used their valence electron counts to begin reasoning about WHY atoms might bond: Al has 3 valence electrons it 'wants' to lose; Cl has 7 and 'wants' to gain one; Na has 1 and 'wants' to lose it; C has 4 and could share. Teachers should introduce —

	valence electrons for each element and recorded Lewis dot symbols. The teacher used cold-calling (minimum 4 students per question) with 10–15 seconds of wait time before accepting answers, ensuring broad participation. Students then used a structured prediction template to forecast: (a) what type of bonding each substance would exhibit, and (b) which physical properties they would expect as a result.	NGSS Science and Engineering Practice (Planning and Carrying Out Investigations / Constructing Explanations). Students distinguished between predictions based on reasoning (acceptable) and guesses without justification (not acceptable). Pedagogical note: emphasise that a wrong but well-reasoned prediction is scientifically valuable — this normalises revision of ideas, which is central to the unit.	but not yet fully explain — the concept of electron transfer vs. electron sharing as two broad bonding strategies. The DQB is updated with new student questions arising from the prediction activity, such as 'If Al gives away electrons, where do they go?' and 'How does carbon hold itself together if it shares equally?' These questions drive Lessons 3–5.
Lesson 3 Ionic Bonding: Why Does Table Salt Conduct Only When Dissolved?	Students carried out a structured conductivity investigation comparing: (a) solid NaCl crystals, (b) NaCl dissolved in water from Lake Turkana (high salinity — a Kenya-relevant context), (c) distilled water, and (d) sugar solution (as a control non-electrolyte). Results were recorded in a class data table on the board. Students also observed (via a projected video or diagram) the cubic crystal lattice structure of NaCl, noting how ions are locked in fixed positions in the solid state but free to move when dissolved. Pedagogical note: the contrast between solid NaCl (no conductivity) and dissolved NaCl (strong conductivity) is the key empirical anchor for the concept of mobile ions.	Ionic bonds form by the complete transfer of one or more electrons from a metal atom to a non-metal atom, creating oppositely charged ions (cations and anions) that attract each other electrostatically. The resulting giant ionic lattice has: high melting and boiling points (strong electrostatic forces require large energy input to overcome); brittleness (ion layers shift under stress, bringing like charges together, causing repulsion and cleavage); conductivity only in aqueous solution or molten state (ions must be mobile to carry charge). Students should be able to explain why coastal Kenya communities use sea salt (NaCl) and why it behaves as it does in cooking.	Students constructed a written explanation for the salt conductivity data using the claim–evidence–reasoning (CER) framework: Claim — NaCl conducts only when dissolved; Evidence — the conductivity data from the investigation; Reasoning — in solid NaCl, Na^+ and Cl^- ions are fixed in the lattice and cannot move to carry current, but dissolving in water frees the ions to migrate to electrodes. The teacher should use a physical model (students acting as Na^+ and Cl^- ions, locked arm-in-arm, then 'dissolving' by separating) to make the abstract concrete. Update the DQB: students can now partially answer 'Why does salt dissolve and conduct?' Connect back to the sufuria and pencil — 'Those don't dissolve, so how DO they conduct?'
Lesson 4 Valence Electrons, Dot-and-Cross Diagrams, and the Six Bond Types	Students examined a set of six 'mystery substance' cards (each describing physical properties — melting point, conductivity, solubility, hardness) representing the six bond types: ionic (NaCl), metallic (Al), simple covalent molecular (H_2O), giant covalent/atomic (diamond/graphite), polar covalent (HCl), and non-polar covalent (Cl_2). Working in pairs, students sorted the cards by bond type before receiving any instruction, recording their reasoning. The teacher circulated and used targeted questioning: 'What evidence from the card makes you think that?' and 'Which of the kitchen items behaves most like this mystery substance?' Pedagogical note: the card-sort is a formative assessment tool —	Students learned the definitions and distinguishing features of all six bond types: ionic (electron transfer, e.g., coastal NaCl from Mombasa salt flats), covalent (electron sharing, subdivided into polar and non-polar), metallic (delocalised electron sea, e.g., the aluminium sufuria), and giant covalent/atomic structures (e.g., graphite and diamond from carbon). The distinction between bond type and structural type was introduced: two substances can share a bond type (e.g., both NaCl and MgO are ionic) but differ in formula and properties. Students practised Lewis dot symbols for Na, Cl, Al, C, H, and O as the foundation for dot-and-cross diagrams in Lesson 6.	The teacher used a structured comparison table (bond type / what is shared or transferred / example from Kenya context / key property trend) to consolidate the six bond types. Students completed the table collaboratively and then used it to revisit their Lesson 2 predictions, annotating corrections in a different colour pen — a visible record of conceptual change. The DQB is updated: students can now name the bond type for each kitchen item and are ready to ask deeper questions about structure (e.g., 'If Al has a sea of electrons, why doesn't it dissolve like salt?'). Pedagogical note: the structured comparison table doubles as a revision resource for the end-of-unit assessment.

	collect reasoning notes to identify persistent misconceptions before Lesson 5.		
Lesson 5 Metallic Bonding and the Aluminium Sufuria: The Electron Sea Model	Students investigated properties of the aluminium sufuria and a copper wire (both metallic) through a set of physical tests: electrical conductivity, thermal conductivity (warming one end and timing heat travel), malleability (hammering a thin Al foil without shattering), and ductility (stretching a wire). Results were compared against the ionic (NaCl) and covalent (graphite) data gathered in earlier lessons. Students plotted the three substances on a class comparison chart: melting point vs. conductivity. Pedagogical note: the thermal conductivity test connects directly to the everyday experience of burning one's hand on an aluminium sufuria handle — a powerful Kenya kitchen context that makes the phenomenon personally meaningful.	Metallic bonding involves a lattice of positive metal ions (cations) surrounded by a 'sea' or 'cloud' of delocalised electrons that are free to move throughout the entire structure. This model explains: high electrical conductivity (delocalised electrons move freely under a potential difference); high thermal conductivity (mobile electrons transfer kinetic energy rapidly); malleability and ductility (ion layers slide past each other without breaking bonds, because the electron sea continuously re-bonds them); high melting points (strong electrostatic attraction between cations and the electron sea requires large energy to overcome, though generally lower than giant ionic/covalent structures for lighter metals). Students connect this to why Kenyan jua kali artisans can hammer and shape aluminium sufuria without the metal cracking.	Students constructed a labelled diagram of the metallic bonding model for aluminium, showing Al^{3+} cations in a regular lattice and the delocalised electron sea. They annotated the diagram to explain each physical property tested. The CER framework was applied: 'Al conducts heat because... the evidence is... the reasoning using the electron sea model is...' The teacher should explicitly address the common misconception that 'metals conduct because they are hard' — hardness and conductivity have different structural explanations. Update DQB: students can now fully explain the sufuria's behaviour. The remaining mystery is graphite — a non-metal that conducts. 'How can carbon form an electron sea without being a metal?'
Lesson 6 Seeing the Invisible: Drawing Ionic and Covalent Bonds with Dot-and-Cross Diagrams	Students practised drawing dot-and-cross diagrams for a progression of molecules and ionic compounds of increasing complexity: H_2 (non-polar covalent), HCl (polar covalent, linking to NaCl comparisons), H_2O (polar covalent, relevant as the solvent in all their conductivity experiments), NaCl (ionic, revisiting Lesson 3 with formal notation), MgO (ionic, used in cooking vessels), and CO_2 (simple covalent, linking to photosynthesis in Sub-Strand 3). Students peer-marked using a provided answer key, identifying the most common error (mixing up dots and crosses from the same atom, or misplacing lone pairs). Pedagogical note: dot-and-cross diagrams are a high-frequency exam skill — budget adequate practice time and use peer-marking to build self-assessment capacity.	Ionic bonding involves complete electron transfer from a metal to a non-metal, producing oppositely charged ions (e.g., Na^+ and Cl^- in table salt from Kenyan coastal markets). Covalent bonding involves the sharing of electron pairs between non-metal atoms to achieve full outer shells. Dot-and-cross diagrams represent these electron arrangements: in ionic diagrams, the transferred electron is shown in the new ion's outer shell (using a cross to indicate its origin); in covalent diagrams, shared pairs appear between the two atomic symbols with bonding pairs and lone pairs clearly distinguished. Students can identify bond polarity by comparing electronegativities and can use diagrams to predict the number of bonds an atom will form.	Students explained how the dot-and-cross diagram for NaCl reveals why the compound is ionic (one atom loses, the other gains — no sharing) while the diagram for H_2O reveals why it is covalent (both atoms share). They connected diagram features to physical properties: ionic compounds have full electron transfer → strong lattice → high melting point; simple covalent molecules have shared pairs but weak intermolecular forces → low melting point (e.g., water melts at 0°C while NaCl melts at 801°C). The DQB is updated to note that graphite is covalent but has a very HIGH melting point — prompting the Lesson 7 investigation into giant vs. simple covalent structures.
Lesson 7 Bond Types, Structures	Students analysed a compiled class data set (from Lessons 1, 3, 5) alongside a	Each substance has a structural type determined by its bond type: NaCl forms a	Students constructed a four-quadrant structural type map, placing each kitchen

and Physical Properties: Making Sense of the Data	provided reference table of melting points, conductivities, solubilities, and hardness values for: NaCl, Al, graphite, diamond, H₂O, and CO₂. They constructed a scatter plot of melting point vs. bond type and identified the pattern that giant structures (ionic lattice, metallic lattice, giant covalent) cluster at high melting points while simple molecular structures cluster low. Students then examined physical models or 3-D diagrams of graphite's layered hexagonal structure and diamond's tetrahedral network, noting how carbon forms four covalent bonds in diamond but only three in graphite (leaving one delocalised electron per carbon — the key to graphite's conductivity). Pedagogical note: this lesson is the intellectual climax of the unit — allow extended discussion time and resist rushing to closure.	giant ionic lattice (ions fixed in position → no conductivity in solid state, brittle, high melting point); Al forms a giant metallic lattice (delocalised electrons → high conductivity, malleable, high melting point); graphite forms a giant covalent layered structure (delocalised electrons within layers → conducts, layers slide easily → soft and lubricating, very high melting point); diamond forms a giant covalent network (no delocalised electrons → non-conducting, hardest natural substance, highest melting point). Simple molecular covalent substances (e.g., H₂O, CO₂) have strong covalent bonds within molecules but weak intermolecular forces between molecules → low melting points. The four structural types (giant ionic, metallic, giant covalent, simple molecular) account for virtually all variation in physical properties.	item and reference substance in the correct quadrant and annotating with the bonding explanation for each property. The map serves as a visual synthesis of Lessons 3–7. The teacher used a Socratic discussion (minimum 6 cold-calls, 15 seconds wait time) to draw out the key generalisation: 'It is not the bond type alone, but the structural arrangement of those bonds across the whole substance, that determines physical properties.' Students annotated their Lesson 2 prediction templates with final corrections, completing the visible record of conceptual change. The DQB is nearly complete — most questions have been answered, setting up the Lesson 8 synthesis.
Lesson 8 Final Explanation — Four Structural Types, Jigsaw Presentations and the Complete Story of the Kitchen Items	Students participated in a Jigsaw activity: expert groups each prepared a 3-minute explanation of one structural type (giant ionic / metallic / giant covalent / simple molecular), then regrouped to teach each other. Each student's jigsaw card included: the structural type, a Kenya-context example, a labelled diagram, and a three-property explanation. Students then retrieved their sealed 'prediction letters' from Lesson 1 and read their original ideas alongside their now-complete understanding. The class conducted a final conductivity and property re-test of all three kitchen items, this time narrating the observations in real time using the bonding vocabulary developed across the unit. Pedagogical note: the prediction letter retrieval is a high-impact metacognitive moment — allow 5 minutes of silent individual reflection before class discussion.	Across Lessons 2–8, students learned that the differences between the aluminium sufuria, the graphite pencil, and table salt are explained by four giant structural types and their bonding forces. Bond type (ionic, metallic, covalent) determines structural type (giant ionic lattice, metallic lattice, giant covalent network/layers, simple molecular); structural type determines the nature and number of interactions that must be overcome; the energy required to overcome those interactions determines every measurable physical property — melting point, conductivity, solubility, hardness, and malleability. This causal chain (bond type → structure → particle interactions → properties) is the master explanation for the entire unit and the complete answer to the driving question.	Students wrote individual final explanations answering the driving question in full, using the four-part causal chain as a scaffold: (1) identify the bond type and justify using valence electrons; (2) describe the resulting structural type and draw a labelled diagram; (3) describe the particle-level interactions (electrostatic lattice forces, delocalised electron sea, covalent network/layers, weak intermolecular forces); (4) explain each measured property as a consequence of those interactions. Responses were peer-assessed against a co-constructed success criteria rubric. The teacher closed the unit by returning to the original DQB and ceremonially answering each remaining question, celebrating the class's intellectual journey from kitchen observation to atomic-level explanation. Students updated their Science Notebooks with a final annotated model showing all three kitchen items and their internal bonding structures.

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